

Inferential Statistics

CMPSC 105 – Data Exploration



ALLEGHENY COLLEGE

Goals

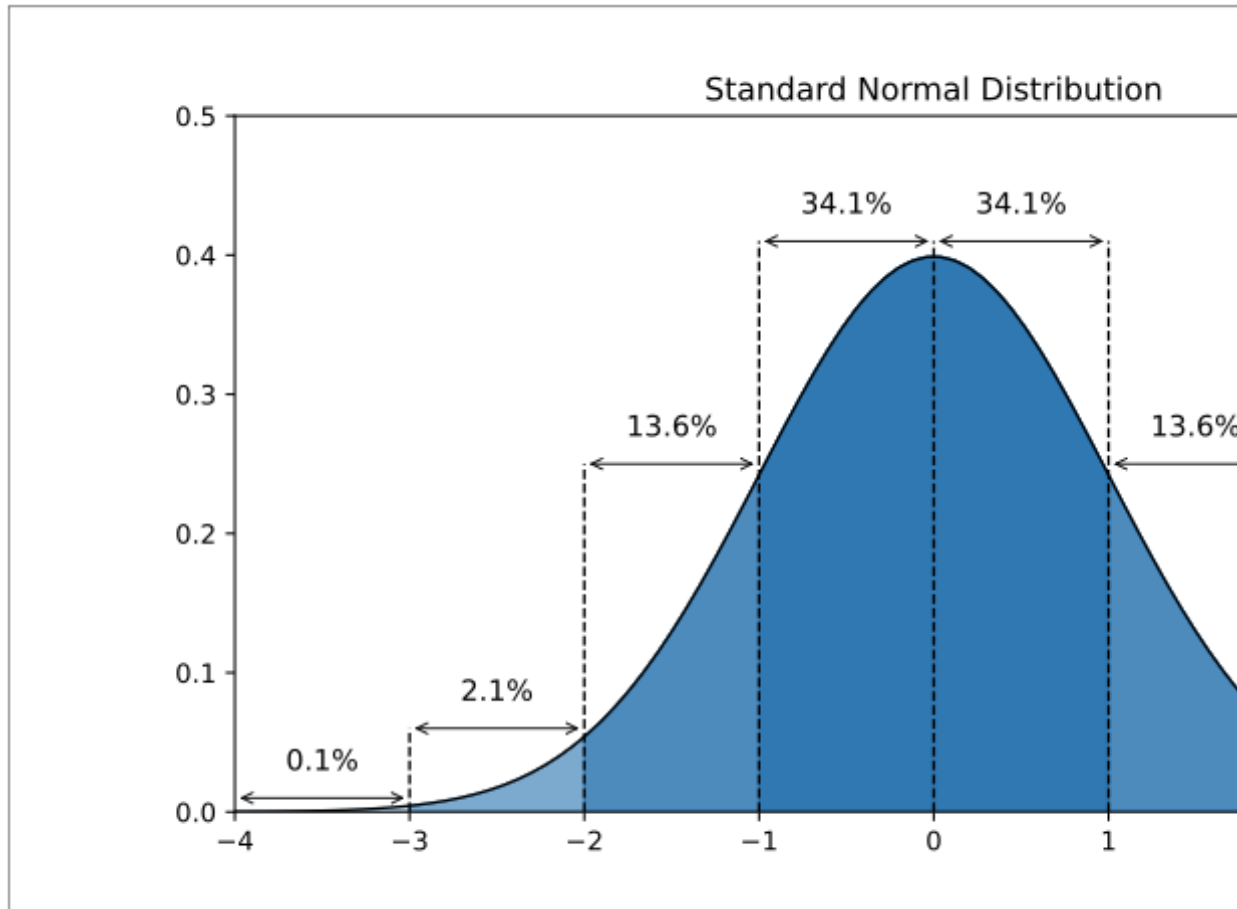
- Standard Normal Distribution
- Single-sample vs multi-sample statistics

Standard Normal Distribution

Standard Normal Distribution

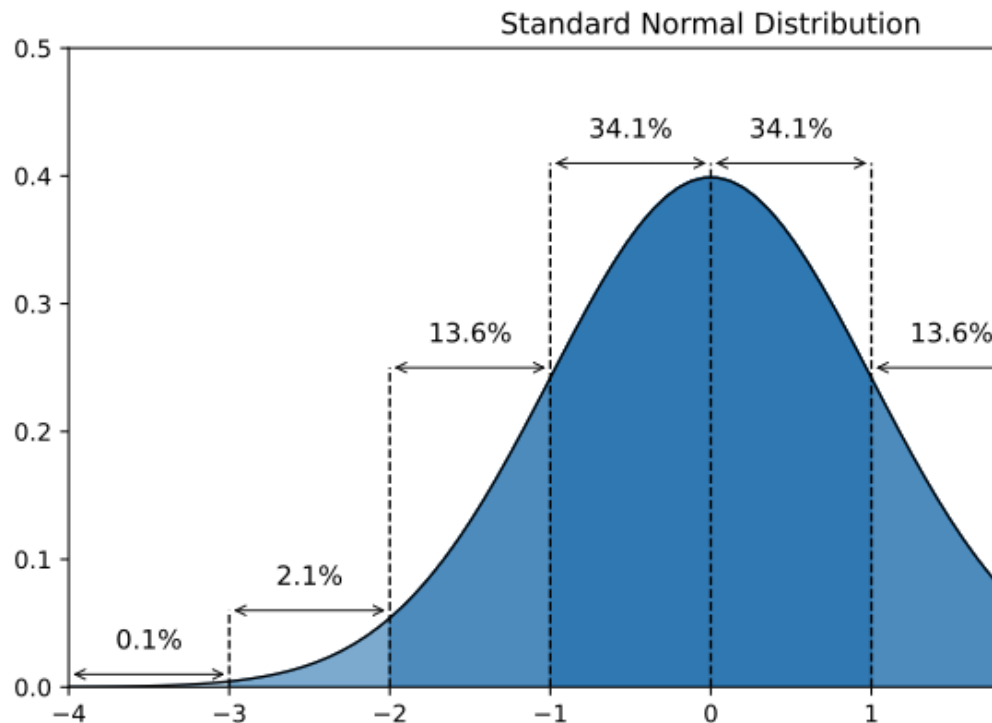
- y axis is proportion
- x axis is z-scores, i.e. how many standard deviations away
- something is from the mean
- Recall $Z_i = \frac{x_i - \bar{x}}{\sigma}$
- area under the curve is unity, 1 aka 100%

Standard Normal Distribution



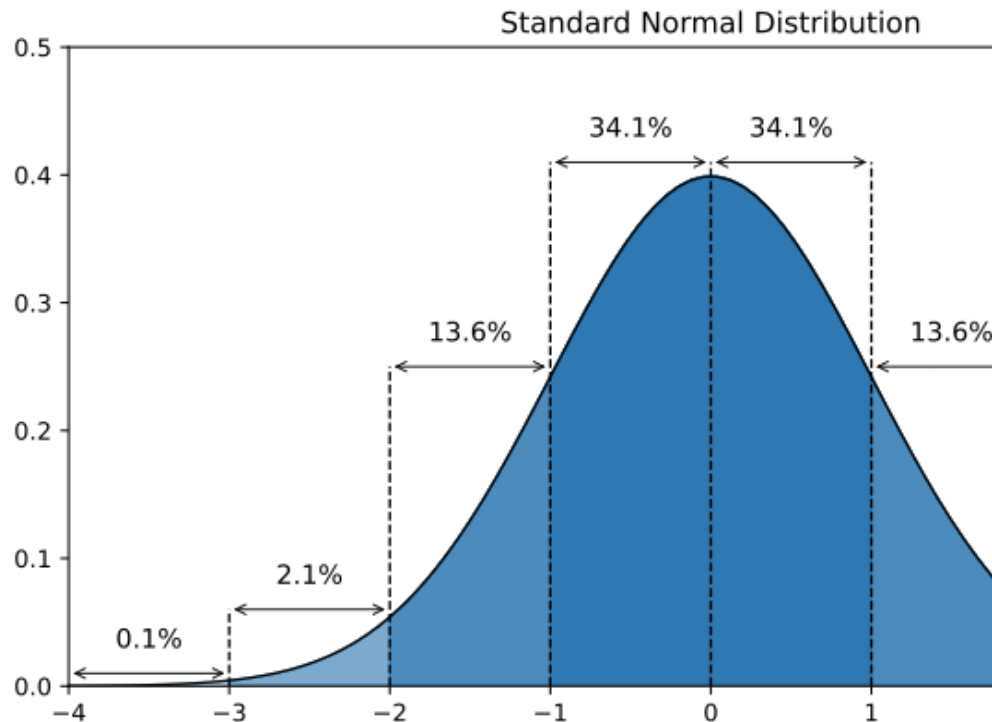
Proportions vs Z scores

- z-scores are on the x axis
- proportions above and below a certain z-score are the area under the curve
- what z-score boundary would separate 2.2% of the area in the right tail?



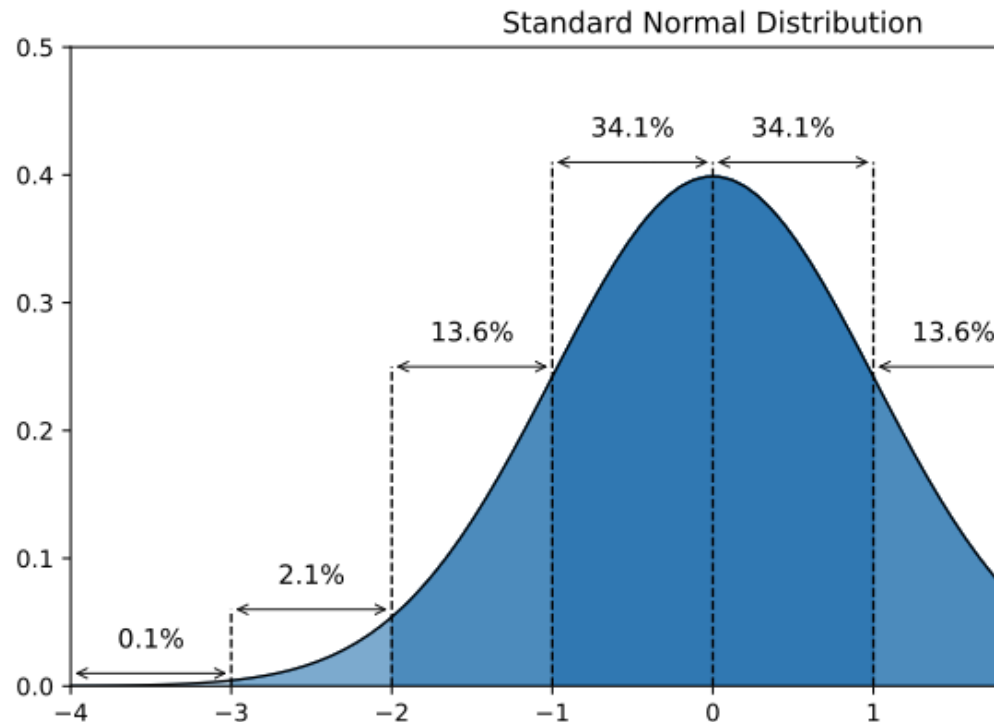
Proportions vs Z scores

- z-scores are on the x axis
- proportions above and below a certain z-score are the area under the curve
- what z-score boundary would separate 15.8% of the area in the left tail?



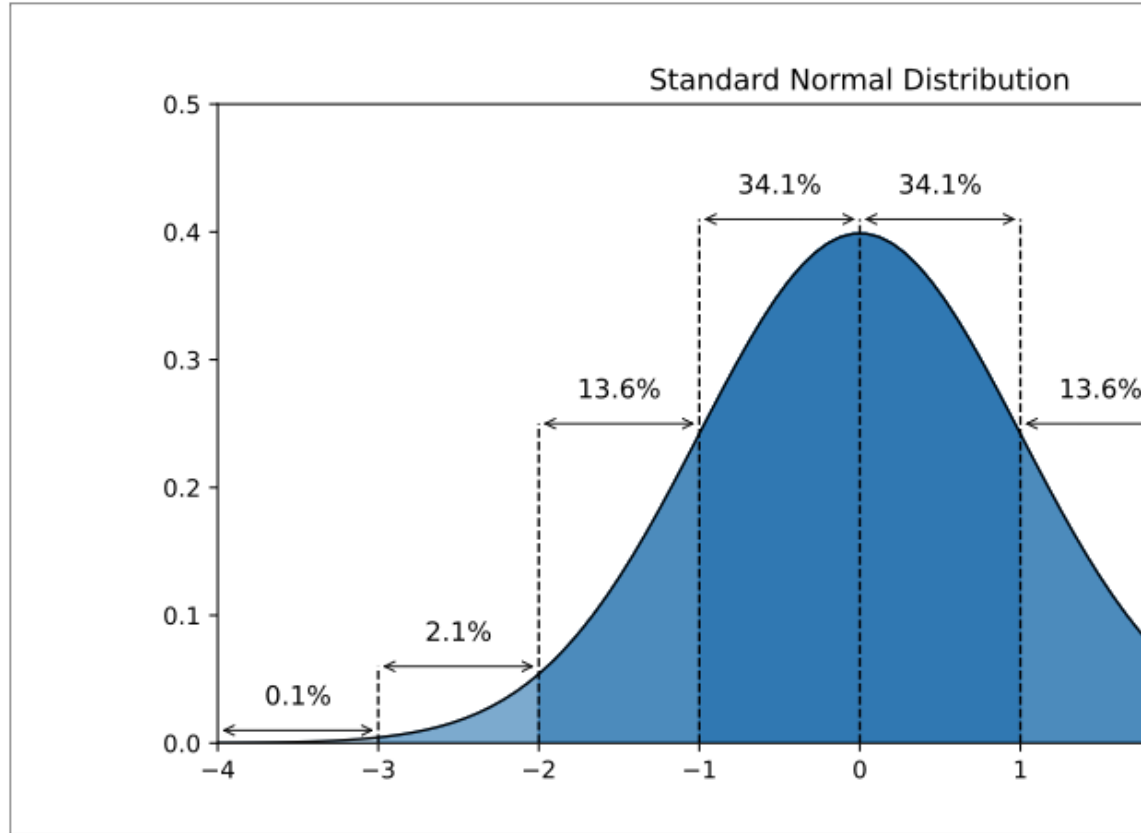
Proportions vs Z scores

- What $\pm$ z-score boundary would separate 4.4% of the area split between the right and left tails?
- Fancy way to ask the same thing: α is .044, $\alpha/2$ is 0.022, what is $Z_{\alpha/2}$?



Proportions vs Z scores

- α is the Greek letter used to talk about proportions of the data out of a total of 1.
- $\alpha/2$ is simple division, splitting the area in half
- $Z_{\alpha/2}$ is asking for a z-score off the x-axis that sections off the requested area.
- if α is .05, what is $Z_{\alpha/2}$?



All Z Values - Unit Normal Table

(A) z	(B) Proportion in body	(C) Proportion in tail	(D) Proportion between mean and z
0.00	.5000	.5000	.0000
0.01	.5040	.4960	.0040
0.02	.5080	.4920	.0080
0.03	.5120	.4880	.0120
...			
0.21	.5832	.4168	.0832
0.22	.5871	.4129	.0871
0.23	.5910	.4090	.0910
0.24	.5948	.4052	.0948
0.25	.5987	.4013	.0987
0.26	.6026	.3974	.1026
0.27	.6064	.3936	.1064
0.28	.6103	.3897	.1103
0.29	.6141	.3859	.1141
0.30	.6179	.3821	.1179
0.31	.6217	.3783	.1217
0.32	.6255	.3745	.1255
0.33	.6293	.3707	.1293
0.34	.6331	.3669	.1331

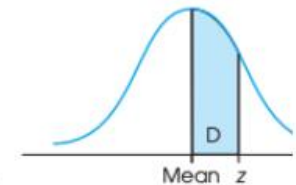
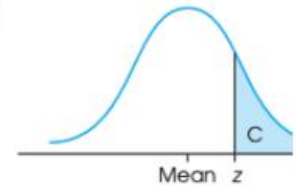
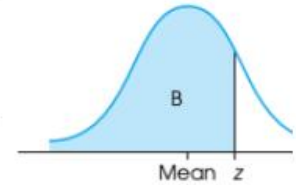


FIGURE 6.6

A portion of the unit normal table. This table lists proportions of the normal distribution corresponding to each z -score value. Column A of the table lists z -scores. Column B lists the proportion in the body of the normal distribution up to the z -score value. Column C lists the proportion of the normal distribution that is located in the tail of the distribution beyond the z -score value. Column D lists the proportion between the mean and the z -score value.

Full Unit Normal Table

Question 1

For a normal distribution, what is the probability of selecting a z-score that is larger than $z = 1.96$?

[Full Unit Normal Table](#)

Z scores and Inferential Statistics

Z scores and the standard normal distribution are essential for inferential statistics.

- z scores make all data comparable
- the standard normal distribution has a fixed area of 1 and can be understood in terms of probability
- we can now ask formalized questions
 - what z-score must some data have for it to only occur 5% of the time or less?
- n.b. 5% is a nice threshold which is commonly accepted as being statistically meaningful

Single Sample versus Multi Sample Statistics

Sample vs Population

A **sample** of the human **population** might be everyone in this classroom, Alden 101. Another **sample** might be everyone in Alden 109. Another **sample** might be everyone in Brooks.

Observations

For each **sample**, there are multiple **observations**.

If I ask everyone in Alden 101 their birth weight, how many **observations** will there be in the **sample**?

Sample Statistics

- Statistics like mean, median, mode, std are descriptive summaries of a sample of a population.
- Statistics that come from a single sample are called a **sample mean**, **sample std**. But...
- The sample mean of birth weight in Alden 101 is likely slightly different than the sample mean of birth weight in Alden 109 right now!
- This means that there exists a distribution of sample means for birth weight!
- The hypothetical distribution of sample means is called the **sampling distribution**.

Sampling Distribution

Sampling Distribution Statistics

- The sampling distribution is conceptually from multiple samples and has stats.
- The mean of the sampling distribution is the expected value of the mean
- the std of the sampling distribution is the standard error of the mean, SEM
- note that we can't actually measure multiple samples, we never see the sampling distribution
- Mathematically and statistically, we can use it

Sampling Distribution Statistics

The central limit theorem states:

For any population with mean μ and standard deviation σ , the distribution of sample means for sample size N will have a mean of μ and a standard deviation of $SEM = \frac{\sigma}{\sqrt{N}}$ and will approach a normal distribution as N approaches infinity, but $N = 30$ is close enough.

Sampling Distribution Shape

Note how the sampling distribution narrows as more observations (N) are used to make the sampling distribution

$$SEM = \frac{\sigma}{\sqrt{N}}$$

Class Coin Exercise

- make a sample with three coin flips (heads = 0, tails = 1).
 - Compute the sample mean with $N=3$
- make a sample with ten coin flips (heads = 0, tails = 1).
 - Compute the sample mean with $N=10$

Standard Error of the Mean

$$SEM = \frac{\sigma}{\sqrt{N}}$$

- recall, SEM is the standard deviation of the sampling distribution
- SEM can be calculated from ONE SAMPLE
- It is desirable to have a small SEM because that would signify that the all the data are close to the true population mean!
- if N of the single sample is very large, the SEM will be small
- if σ of a single sample is small, the SEM will be small

Z scores for one Sample in a Sampling Distribution

- assume the sampling distribution has mean μ and standard deviation $SEM = \frac{\sigma}{\sqrt{N}}$
- assume the single sample mean $\bar{x}$
- recall that z-scored observations from single sample relate each sample to the sample mean $z_i = \frac{x_i - \bar{x}}{\sigma}$
- $z_{sample} = \frac{\mu - \bar{x}}{SEM}$

Question 2

The population distribution of SAT scores is normal with a mean of $\mu = 500$. For a random sample of 25 students ($N = 25$) and a standard deviation of $\sigma = 100$, what is the probability that the sample mean $\bar{x}$ will be greater than 540? Greater than 700?

[Unit Normal Table for standard normal distribution](#)

for $\bar{x} = 540$, $p = 0.0228$, so it is quite unlikely for a sample of 25 students to get an average SAT score higher than 540.

Making Confident Statements

- We can say with a set level of confidence that a single sample mean is within a certain interval, using the following formula:

$$\bar{x} \pm z_{\alpha/2} \times SEM$$

$$SEM = \frac{\sigma}{\sqrt{N}}$$

- $z_{\alpha/2}$ is a fixed number read off of a standard normal distribution when α is chosen.
- α represents the chance that we are wrong, and is area under the curve. It is a proportion of 1, usually, 0.05.

Confidence Intervals

- CIs are chosen by researchers, 95% confidence level is common choice.
- 95% refers to area under a Standard Normal Distribution and corresponds with α of .05
- The Interval that is reported is the z-score that sections off the area

Critical Z-value

- if α is .05, $\alpha/2$ is 0.025 and $z_{\alpha/2}$ is ± 1.96
- if α is .01, $\alpha/2$ is 0.005, what is $z_{\alpha/2}$?

[Full Unit Normal Table](#)

Hypothesis Testing

- A null hypothesis H_0 can be constructed that says, a sample does not have a sample mean $\bar{x}$ that is different from the population mean μ .
- If the sample mean turns out to be unusually far from the population mean, the probability, or p value can be reported. Typically, probabilities or p values that are less than 5% are considered significant.
- The null hypothesis would be rejected and the z-score, p value, mean and standard deviation would be reported.

Critical Z-value

Suppose the average SAT score in the population is 500.

Our null hypothesis is:

$$H_0: \mu = 500$$

We take a sample of 25 students and find:

$$\bar{x} = 540$$

$$\sigma = 100$$

$$N = 25$$

Is 540 far enough from 500 to be considered unusual?